

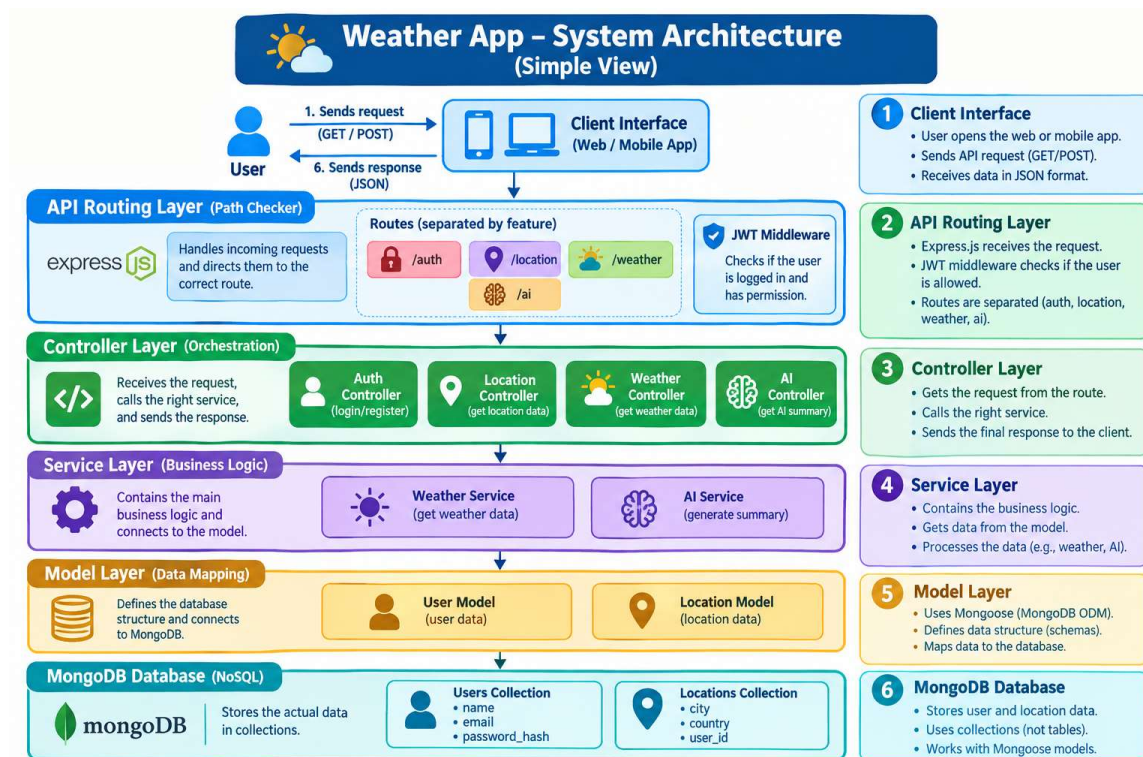
PHASE – 2

Hardware Requirements

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- **Processor:** Intel Core i5 (8th Gen or above) / AMD Ryzen 5 or equivalent.
- **RAM:** 8 GB minimum (16 GB recommended for concurrent instances of MongoDB, processing layers, and execution nodes).
- **Storage:** 1 GB of available disk workspace.

PROJECT ARCHITECTURE



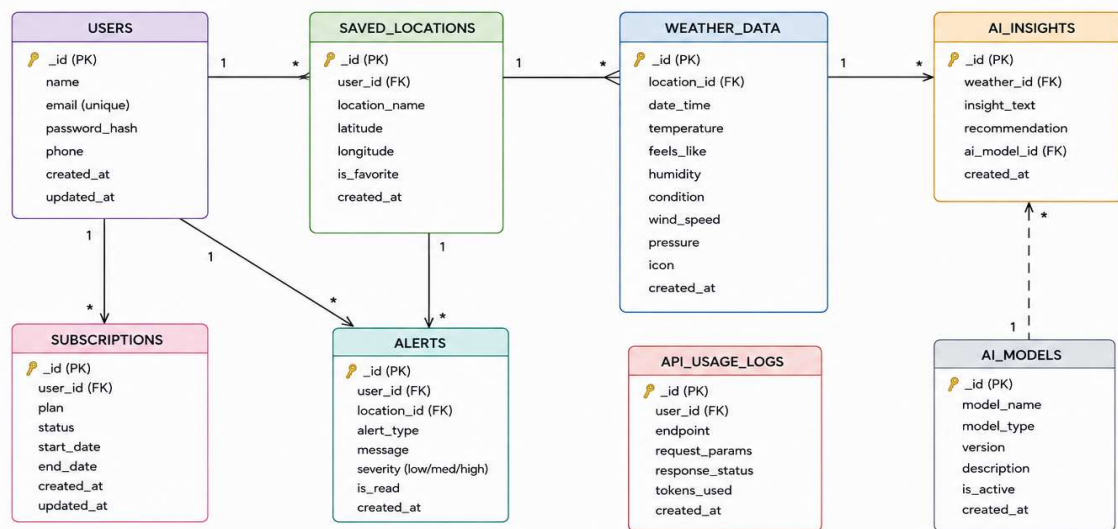
The AI WeatherWise application follows the MVC pattern to decouple business logic from data management. The Controller layer orchestrates weather fetching from OpenWeatherMap and generation of AI insights via Gemini, while the Model layer manages User and Location entities.

Components Definition

- **Interface:** API tools like Postman query distinct route paths to extract JSON entities.

- **Express Server Gateway:** Monitors network ports, maps explicit HTTP methods (GET, POST, PUT, DELETE), parses JSON payloads, and handles CORS headers.
- **Authentication & Role Matrix Middleware:** Extracts tokens from headers, validates user context signatures, and evaluates if the specific role satisfies authorization thresholds for guarded resource paths.
- **Granular Services (Logic Modules):** Enforces isolated domain rules for blog lifecycles (Draft, Pending Approval, Scheduled, Published), comment queues, and system reporting indicators.
- **Database Interface Layer:** Mongoose translates operational application logic into performant native MongoDB queries.
- **AI Service Integration:** Leverages @google/generative-ai to generate weather summaries and personalized recommendations based on live temperature and conditions.

Entity-Relationship (ER) Diagram Description



The data architecture manages User profiles and their associated favorite Locations through logical references:

1. User Entity

- `_id` (Primary Key - ObjectId)
- `name` (String, Required)
- `email` (String, Required, Unique)
- `password` (String, Hashed, Required)
- `role` (String, Default: "reader")

2. Location Entity

- `_id` (Primary Key - ObjectId)
- `user` (Foreign Key referencing User Collection)
- `city` (String, Required)
- `country` (String, Required)
- `createdAt` (Date, Default: Now)